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Effect of the appropriateness of sound environment on urban soundscape assessment

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ABSTRACT

Sound environment is a critical factor to be considered in the development of sustainable and healthy urban communities and cities. We investigated the effect of the appropriateness of the sound environment on urban soundscape perception through individuals' subjective evaluation in a laboratory condition. Our aim was to propose design guidelines to enhance urban soundscape quality. Virtual reality technology was used to simulate a realistic soundscape: visual information was generated via a head-mounted display, and spatial audio information was provided via headphone-based first-order ambisonic head-tracking technology. The appropriateness and perceived degree of sound sources, perceived visual elements, and soundscape quality (such as satisfaction, desire to stay, and comfort) were evaluated with 30 participants in two experimental setups: a visual environment and an audio-visual environment. The results revealed that the appropriateness of the sound sources interacted with individuals' perception of visual elements, and "traffic" sounds and "birdsongs" were critical factors that influenced participants' initial perception of urban soundscape quality. Further, "human" sounds originating from human activity were related to the "comfort" aspects of soundscape quality. The findings and soundscape design guidelines will be useful for the planning and development of urban soundscapes.

1. Introduction

Sound environment is an important factor to be considered in building sustainable and healthy urban communities and cities. This has led to the development of soundscape concepts that are based on humans' perception of the sound environment [1–3]. The International Organization for Standardization [4] defines a soundscape as "an acoustic environment that is perceived or experienced and/or understood by a person or people in context." The earliest soundscape studies were conducted in a variety of urban environments such as parks, open spaces, and streets; they used various methods to investigate the relationships between soundscape indicators (e.g., acoustic indices and context) and soundscape descriptors (e.g., comfort, tranquility, capacity for restoration, and soundscape quality) [5–10].

To interpret individuals' perception of soundscape, it is necessary to consider the contexts as formulated by several complex factors, including non-acoustic factors (e.g., the function or type of a natural or constructed environment such as a park, square, or street), social factors (e.g., individual experience, lifestyle, or culture), and acoustic factors [11,12]. These urban environment contexts are described as the "circumstances that form the setting for an event, statement, or idea, based on which it can be fully understood and assessed [13]." Several researchers have investigated the relationship between the context and soundscape perception, revealing that soundscape perception is influenced by a combination of person, place, person–place interaction, activity, and environmental experiences [5,11]. Previous studies have revealed that, if a given space has a context, individuals' expectation from the context may significantly influence their perception of the space [14–16]. Kang [16] proposed that, when individuals enter a different place, they expect a change in the sound environment. Therefore, expectations, such as the expected type of sound source in a space, may contribute to the context [17]. Accordingly, previous studies have revealed that the expectation from a space can be used to interpret the competence of a soundscape [18,19]. Bruce and Davies [15] investigated the effect of sound environment expectation on soundscape perception while accounting for other relevant factors.

"Appropriateness"—whether a soundscape is appropriate for a place—was proposed by Brown, Kang, and Gjestland [20] for urban planning and design. Davies, Bruce, and Murphy [21] found that, when designing urban soundscapes, participants placed more importance on

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their expectation of the urban environment than on their individual preferences. Axlsson [22] also found that appropriateness, as well as perceived affective quality such as pleasantness and eventfulness, could be an important independent descriptor for assessing soundscape quality. Such "appropriateness" has considerable importance in terms of soundscape congruency. The congruency of soundscapes and landscapes enhances an individual's experiences of presence and enjoyment in the space [23,24]. It is known that such congruency positively affects an individual's experiences (tourism experience, walking pace, etc.) [25,26]. In addition, congruency of sound affects attentional control for visual ambiguity [27]. Furthermore, it was revealed that increased congruence of individual expectations regarding sound and soundscapes in urban parks led to greater acceptability of sounds [28]. As mentioned previously [5,15], it is clear that "appropriateness" is a potential descriptor for the characterization of soundscapes; however, there is insufficient background research that validates its applicability. Although some studies [29,30] assessed appropriateness in evaluating soundscapes, they only used it as a supplementary indicator for interpreting preferences for spaces; therefore, it is difficult to apply these findings to urban design and planning processes. Additionally, there has been no study on the variation of appropriateness for individual sound sources according to different urban contexts or on how the congruency between appropriateness and perception specifically affects soundscape quality evaluation.

In contrast, a multi-sensory design approach is critical considering holistic perception in analyzing urban environments; therefore, several studies have investigated the effects of audio-visual interactions on soundscape perception [31–38]. Jeon and Jo [38] derived the perception dimension of soundscape and landscape based on the presence of audio-visual information and proposed a perception model. However, they failed to consider the differences in the assessment of appropriateness levels represented by an individual's experiences or knowledge. As the effect of the audio-visual interaction on the environment assessment is obvious [39], in this context, a comprehensive analysis of appropriateness on sound environments is required considering the visually perceived factors in urban environments. Moreover, concerning the assessment methodology of urban environments, most previous soundscape enhancement studies were conducted in the field and involved processes such as soundwalks and narrative interviews, which were used to ensure full ecological validity [40,41]. Recent studies applied virtual reality (VR) technology to numerous sound environment evaluation tests to provide more immersive experiences for their participating subjects in limited laboratory environments [42–44]. Various studies have suggested audio-visual methods for implementing VR environments that closely simulate actual active urban spaces and have verified the effectiveness of applying VR technology to assess soundscapes [24,45–49]. The results of some studies revealed that VR shows sufficiently high congruency in response to the actual environment and perception of audio-visual stimuli [50,51]. Studies that compared headphones and speaker-based VR systems have shown that for sound- or environment-related attributes, the use of VR technology can elicit responses similar to those of the real environment [24]. Therefore, even though VR technology cannot fully simulate various physical senses such as odors and lighting, it can be sufficiently applied to the evaluation of the audio-visual perception of an urban environment.

Thus, our purpose was to investigate the appropriateness of the sound environment on individuals' perception of soundscape quality and to propose design guidelines for the enhancement of urban soundscape quality. For this purpose, two VR experimental setups were designed, and 30 participants completed a subjective evaluation. We compared the appropriateness of the sound environment with the perceived dominance of sound sources in the urban environment. For an in-depth interpretation of soundscape perception, the audio-visual interaction was also analyzed. Finally, we analyzed the relationship between soundscape quality perception and the appropriateness of the sound environment to propose urban soundscape design guidelines regarding the appropriateness of the sound environment.

2. Material and methods

2.1. Stimuli

2.1.1. Site selection

To investigate the effect of the appropriateness of a sound environment on the perceived quality of urban soundscape, six different sites in Seoul were selected (Fig. 1). Each site included various land uses such as roads and open or green spaces as well as multiple sound sources such as traffic, human sounds, and birdsongs. Table 1 presents the functions and spatial characteristics of the considered sites. Site A was located close to a road and shops and was characterized by heavy vehicular traffic and numerous pedestrians. Sites B and C were public squares located far from roads. However, site B served as a pedestrian pathway and was lined with trees, whereas site C was surrounded by buildings and lacked significant pedestrian traffic. Sites D and E were green spaces. However, site D was adjacent to a road and contained recreational facilities such as benches, while site E was far from the road and densely populated by trees, functioning more as a pedestrian passage. Site F was a recreational space located at a higher altitude than the surrounding areas. It provided panoramic views of cafes and shops on one side and a recreational area with benches and tables on the other side.

2.1.2. Audio-visual recording

To implement the VR soundscape assessment in a laboratory, audio-visual information was collected from the six selected sites between 10 a.m. and 2 p.m. on a day in September 2018. An 8 K ultra-high-definition spherical panoramic camera (Insta 360 pro, Insta 360, China) with a temporal resolution of 80 fps and bit rate of 95 Mbps was used to acquire high-quality images over an angular range of 360°. An ambisonic microphone (Soundfield SPS 200, Soundfield Ltd., England) and a portable sound recorder (MixPre-6, Sound Devices, USA) were used to simultaneously record audio information in A-format first-order ambisonics (FOA) [38,49,52]. Both visual and audio data were measured for 3 min according to the minimum recommendations (ISO 12913–2 [53]), and the height of the camera and microphone tripod was adjusted to 1.6 m according to the standard. When the sound source meant for assessment was activated in the laboratory environment, A-weighted sound pressure levels (SPLs) (L_{Aeq}) were measured using a 0.5-in microphone (GRAS 46AE, GRAS Sound and Vibration, Denmark) and a portable sound level meter (AS-70, RION, Japan) was used to calibrate the sound pressure to the same level as that at the actual site. Clapping was used to signal the onset of measurement and synchronize the video and audio information.

The physical properties of the soundscape at each assessment site were examined based on indicators related to the sound strength, spectral contents, and temporal structure, as detailed in Table 1. The A-weighted SPL (L_{Aeq}) was calculated to determine the sound strength characteristics. Additionally, the difference between the A-weighted and C-weighted SPL, $L_{Ceq–Aeq}$, which is an indication of the relatively low-frequency characteristics of the sound spectral contents, was analyzed. The $L_{A10–A90}$ values were calculated to investigate the temporal variability of the sound environment. The average L_{Aeq} values of the assessment sites were distributed within 57.2–79.4 dBA, and the SPLs were high in areas close to roads. In addition, the differences between the maximum and minimum values of $L_{A10–A90}$ and $L_{Ceq–Aeq}$ for the different sites exceeded 10 dB. This indicated that the values accounted for the sound variation range of a typical urban area.

2.1.3. VR environment reproduction

The images acquired by the six-channel 360° camera were stitched using Insta360 (version 2.1.0) stitching software (Insta360stitcher, Insta360, China) during preprocessing; thereafter, they were

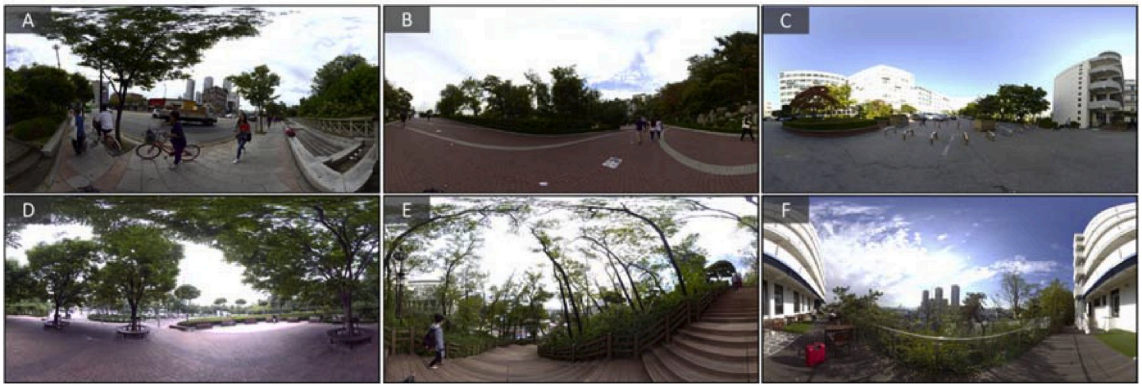

Fig. 1. Equirectangular panoramic images obtained from spherical video recordings at the selected assessment site.

Table 1
Details of the assessment sites.

Site	Function and description	L_{Aeq}	$L_{A10-A90}$	$L_{Ceq-Aeq}$
A	Busy road with shops *Description: busy road, considerable traffic, and human activity near bus and subway stations*	79.4	12.5	16.5
B	Public square *Description: pedestrian square, away from traffic, and with trees on either side*	65.7	11.6	6.6
C	Public square *Description: square surrounded by buildings, small trees, and numerous paved areas*	57.2	11.4	9.3
D	Green space *Description: small urban park close to a road with public seating*	68.4	10.1	13.4
E	Green space *Description: green area with wooden stairs away from vehicular traffic*	64.8	15.8	5.7
F	Recreation space *Description: expanded view, elevated location overlooking retail shops on one side and trees on the other side*	65.1	5.6	7.7

interconnected via a head-mounted display (HMD). Hong and colleagues [24] reported that the overall soundscape quality, sound-related attributes, and environment-related spatial attributes experienced by participants were similar to those in the field when head-tracking technology was applied to headphone-based FOA. Therefore, in this study, to produce headphone-based spatial sound, the sounds recorded in A-format FOA were converted into B-format FOA and then down-mixed to a binaural track using Spatial Audio API (Google VR, USA), embedded in the Unity engine software version 2018. In this case, a head tracker was used to implement the real-time directivity based on participants' head rotation. This enhanced the sense of immersion and realism for the assessor in the virtual audio environment. To calibrate the SPL of each sound source for each assessment site in the laboratory environment, six sound sources were reproduced by VR via headphones (Sennheiser HD-650) in a semi-anechoic chamber, which allowed for sound recording using a head and torso simulator (Type 4100, Brüel & Kjær, Denmark). The A-weighted equivalent continuous sound levels ($L_{\mathrm{Aeq,\ 3\ min}}$) of the different sources were calculated based on the recordings. The respective SPLs were then calibrated using Adobe Audition software (version 1.5, Adobe, USA) and subsequently set to the $L_{\mathrm{Aeq,\ 3\ min}}$ values of the corresponding sound sources, which were recorded by the sound-pressure-calibrated microphone at the actual sites [38,45].

2.2. Experimental design

2.2.1. Setups

Our main objective was to investigate the influence of the appropriateness of a sound environment on the quality assessment of an urban soundscape. Two different assessment setups were constructed for this purpose: (1) a visual environment and (2) an audio-visual environment. The visual environment was generated by providing only visual information through the HMD without any audio information. It was used to investigate individuals' expectations for the sound environment, especially sound sources, based on their prior experiences and various non-acoustic factors such as preference, identity, and so on. The audio-visual environment was generated by providing audio and visual information, which was used to examine the perceived appropriateness of sound sources at the assessment sites. In other words, the visual information of both environments was provided in the same setups, and the only difference was the presence and absence of sound in setups 1 and 2, respectively. All experiments were performed in the same semi-anechoic chamber, with background noise maintained at a low level of ~25 dBA. An open-type headphone (HD-650, Sennheiser, Germany) was used to supply the audio information, while the visual information was supplied via an HMD (VIVE Pro, HTC, China).

2.2.2. Questionnaire

The questionnaire varied based on the environment setups. As shown in Table 2, the degree of appropriateness of sound sources in the visual environment setups (1-1) and the perceived degree of sound sources in the audio-visual environment (1–2) were evaluated. Additionally, perceived visual elements (2) and soundscape quality (3) were assessed in both environment setups.

In the visual environment setups, the following question (1-1) was asked to investigate the expectation of sound sources at each assessment

Table 2
Questionnaire details.

Questions	Likert scale (−2 to 2)
1-1. Visual environment (Day 1) - To what extent do you presently think it is appropriate to hear the following types of sounds: traffic, human, and birdsongs?	Not appropriate at all/ definitely appropriate
1-2. Audio-visual environment (Day 2) - To what extent do you currently hear the following types of sounds: traffic, human, and birdsongs?	Did not hear at all/dominant sounds
2. Visual/audio-visual environment (Day 1 and 2) - To what extent do you currently see the following types of visual elements: vehicle, building, road, open space, vegetation, people, and sky?	Did not see at all/completely dominant
3. Visual/audio-visual environment (Day 1 and 2) - To what extent are you satisfied with the present environment? - To what extent do you want to stay in the present environment? - To what extent do you feel comfortable in the present environment?	Very bad/very good

site: "To what extent do you presently think it is appropriate to hear the following types of sounds?" Participants responded on a five-point Likert scale ranging from −2 to 2. In the audio-visual environment, the following question (1–2) was asked to examine the dominant sound source at each assessment site: "To what extent do you presently hear the following types of sounds?" Participants responded on a five-point Likert scale ranging from −2 to 2. Brown, Gjestland, and Dubois [54] classified the sounds in an urban acoustic environment into the following categories: sounds generated by human activities, sounds generated by facilities, and other biophonic or geophonic sounds. In other previous soundscape studies [55,56], five categories of sound sources were proposed: vehicular traffic, fans, other noises, human sounds, and natural sounds. In addition, other categorizations of "traffic, " "human" sounds, "birdsongs," "water," and "music" have been extensively employed in recent works [35,36,38,57]. The sounds of water and music were not present at the examined sites; therefore, only the other three categories of sounds—traffic (e.g., cars, buses, trains, airplanes), human (e.g., conversation, laughter, children at play, footsteps), and birdsongs—were evaluated in the present study.

The following question (2) was asked in both the experimental setups regarding the dominant visual elements at each assessment site: "To what extent do you currently see the following types of visual elements?" Participants responded on a five-point Likert scale (−2 to 2). Based on previous studies, we considered that visual environment quality plays a key role in the perception of soundscape quality [6,34,38,58,59]. Furthermore, in several previous works on landscape assessment, visual components were categorized as "vehicle," "building," "road," "open space," "green space," "people," and "sky" [6,31,35,38,60].

Finally, the following corresponding questions (3) were asked: "To what extent are you satisfied with the present environment?", "To what extent do you want to stay in the present environment?", and "To what extent do you feel comfortable in the present environment?" Participants responded using a five-point Likert scale (−2 to 2). Regarding the urban soundscape quality, among the various direct and indirect outcomes of the soundscape preference assessments conducted by Brown et al. (2011), three related items were considered in the present study: "satisfaction," "desire to stay," and "comfort." We considered that there was a limit to assessing the perceived soundscape quality with only one indicator; thus, we examined all three.

2.3. Procedure

Thirty participants (men = 20, women = 10) participated in the soundscape assessments; this number was the same as in previous studies [6,38] and satisfied the minimum criteria (ISO 12913–2). To reduce the variability of individual preferences, which could be influenced by experience and knowledge, all participants were recruited from the same university and had previously visited all six sites. Before the assessments, all participants provided informed written consent for their participation and were compensated accordingly. Participants' ages ranged 22–29 years (mean age = 24.80 ± 1.97 years). Before the tests, an audiometer (AA-77, RION, Japan) was used to assess participants' auditory abilities to confirm that they had normal hearing.

Each participant completed the soundscape assessments in the two considered environments: the visual environment on the first day and the audio-visual environment on the second day. Completing the experiments on separate days—day 1 and day 2—according to the experimental setups instead of on a single day enabled controlling the memory effects of the visual environment (day 1). In addition, allowing the same participants to participate in both experiments eliminated the interpersonal variability [61]. Furthermore, to control possible biases in the participants' responses and attention, we did not inform them prior to the experiment regarding whether there would be any audio or visual information. The participants were only requested to respond to the survey questionnaires on the environment presented to them. The six different stimuli representing assessment sites A–F were sequentially presented to participants. The length of the stimuli was 3 min in the audio-visual environment, and the stimuli were repeated at the participants' request. All participants were presented with VR stimuli while sitting in a set position at a particular location. Although they were not allowed to move their positions within the VR environment, they were allowed to turn their heads to freely examine their surroundings. Simultaneously, a stereophonic sound effect was applied in real time as they rotated their heads. A questionnaire graphical user interface was developed and implemented in the VR device, and participants could respond to the questionnaire through the HMD controller. In addition, to eliminate physical discomfort to participants, who were required to wear the HMD throughout the experiment, the duration of the assessment of each environment was limited to less than 30 min. Participants were allowed sufficient time to rest when they requested it. Participants were to respond to all survey items in each assessment point; thus, six response results were obtained from each participant (N = 360 data samples: 2 visual environments × 30 participants × 6 sites).

2.4. Statistical analysis

SPSS (version 25, IBM, USA) software was used for statistical analyses. Kolmogorov–Smirnov and Shapiro–Wilk tests were used to analyze participants' questionnaire responses. The normality criterion was not satisfied when the significance level was less than 0.05; therefore, nonparametric statistics were applied, and the Wilcoxon signed rank test (Z test) was employed. For an in-depth analysis of participants' responses, first, the mean response data of all assessment sites were compared before and after the audio information provision; then, the individual assessment sites were compared. The rank correlation (Spearman's rho coefficient, r_s) was used to examine the relationship between the soundscape quality and perceived audio-visual elements. We measured the difference (Δ, the average rating score in the visual environment – the average rating score in the audio-visual environment); if the difference value was small, the congruency between the two experiment setups was good. We analyzed the relationship between this difference in perception and the change in the assessment of the soundscape quality according to the different setups.

3. Results

3.1. Sound sources

Fig. 2 shows the degree of appropriateness of sound sources and the degree of perceived appropriateness of various sound sources based on the different experimental setups. The degree of appropriateness indicates what the participants expect to hear at the various sites in the visual environment setup in which there is no auditory information. The Spearman correlation coefficients (r_s) of the two different experiment setups were analyzed to investigate the congruency of subjective response between the degree of expectation and the degree of perception of the sound source. For each type of sound source (e.g., "traffic" (r_s = 0.59, $p < .01$), "human" ($r_s = 0.38$, $p < .01$), or "birdsongs" ($r_s = 0.40$, $p < .01$)), the quantitative linear correlation between participants' expectation and actual perception was significant.

The Wilcoxon signed rank test was used to determine the significance of the inappropriateness between the expectation and the actual perception of the sound source according to the assessment environment (see Fig. 2). For "human" sounds and "birdsongs," the degree of actual perception at the assessment site with audio-visual information was lower than the degree of appropriateness of the sound sources in the visual environment (0.76 and 1.16, respectively; $p < .01$), but not for "traffic" sounds. The results of the Wilcoxon signed rank test for individual sites are also shown in Fig. 2. For the "traffic" sounds, there were significant differences between certain sites (A and D), which were located close to roads, and site F, which was an elevated area.

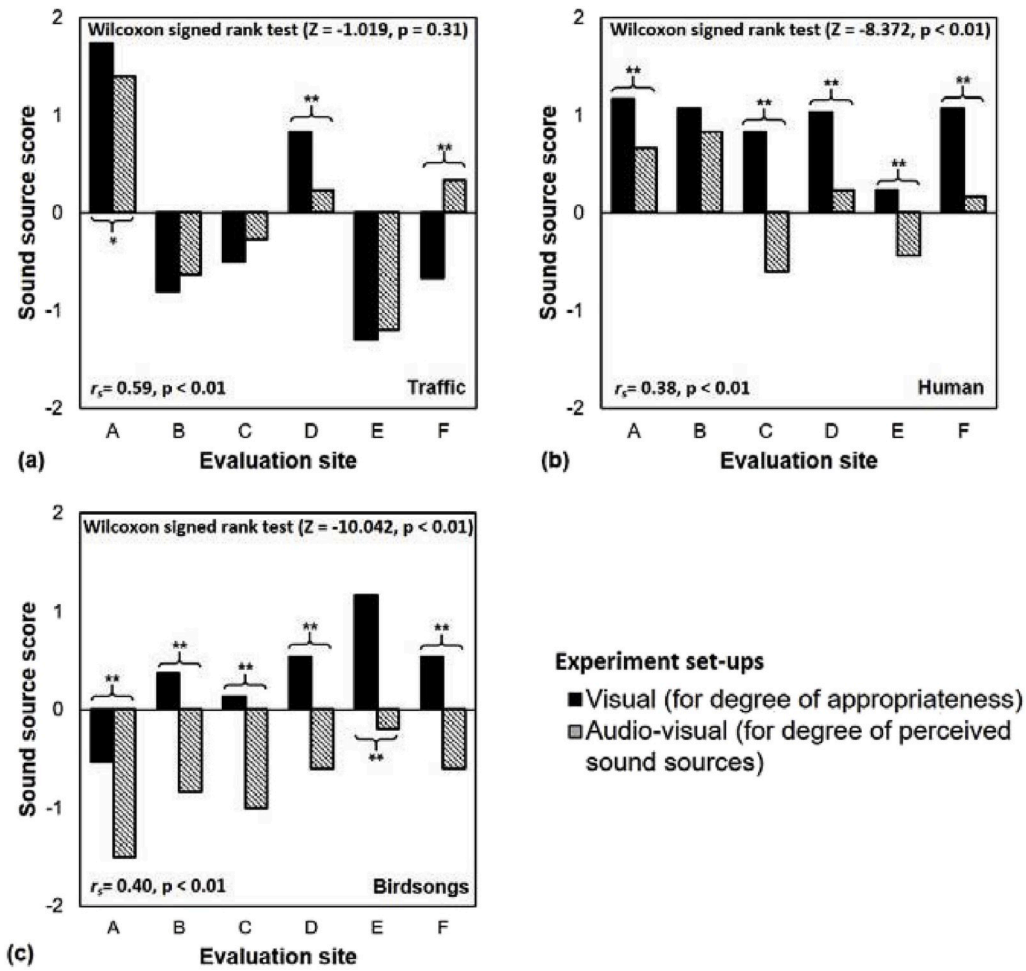

Fig. 2. Mean rating scores with the degree of appropriateness of sound sources and actual perceived sound sources according to the different experimental setups at the assessment sites: (a) traffic, (b) human, and (c) birdsongs; $*p < .05$, $**p < .01$.

Concerning "human" sound, significant differences were observed for the averaged data of assessment sites, with the exception of site B. "Birdsongs" also yielded significant differences for all assessment sites. Moreover, "birdsongs" demonstrated the largest difference between the degrees of appropriateness and actual perception. In other words, an evident inappropriateness was observed between the expected degree of participants' evaluations of sound sources as appropriate for a particular space in the visual environment and the degree of sound actually perceived in urban spaces.

3.2. Perceived visual elements

Fig. 3 shows the degrees of perception for the dominant visual elements at each assessment site in the visual and audio-visual environments. The most perceived visual elements differed according to the assessment site, and even in spaces that performed similar functions, perception of visual elements differed according to the combination of surrounding contexts. The Spearman correlation coefficients (r_s) of the two different experiment setups ranged from 0.43 to 0.72, indicating that the congruency of participants' responses was quite high and significant ($p < .01$).

The results of the Wilcoxon signed rank test conducted on the total assessment results are presented in Fig. 3. With the exception of "road" and "open space," the perception differences regarding "vehicle," "building," "green space," "people," and "sky" according to the different experiment setups were significant ($p < .01$). "Vehicle" yielded more dominant perception outcomes in the audio-visual information (mean = −0.08) than did only the visual environment (mean = −0.32), and the other foregoing visual elements ("building," "green space," "people," and "sky") were less perceived in the audio-visual environment than those in the visual environment (mean = 0.28, 0.24, 0.40, and 0.38, respectively; $p < .01$). Hence, even in the same space, the perception of visual elements may differ when participants' expectations from the sound environment and the actual degree of perceived sound sources are incongruent.

The statistical significance of the difference in visual perception at each site according to the experiment setups was also analyzed (see Fig. 3). The sites (B, C, and F) where transportation was not visible in the view from the space yielded significant perception differences for "vehicle." This suggested that the input of sound information improved the perception of "traffic" sounds, resulting in increased visual focus on "vehicle." It is noteworthy that when sound information was provided, the perceptions for "people" were significantly lower for most assessment sites. The reason for this can be the input of audio information, which diminished participants' visual attention on human presence in the space. Generally, the perception of the visual elements varied according to the presence or absence of audio information. This shows that the appropriateness of sound sources significantly influenced the perception of urban environments, and this is closely related to the presence of visual elements.

3.3. Soundscape quality

For the soundscape quality at the various assessment sites, as shown

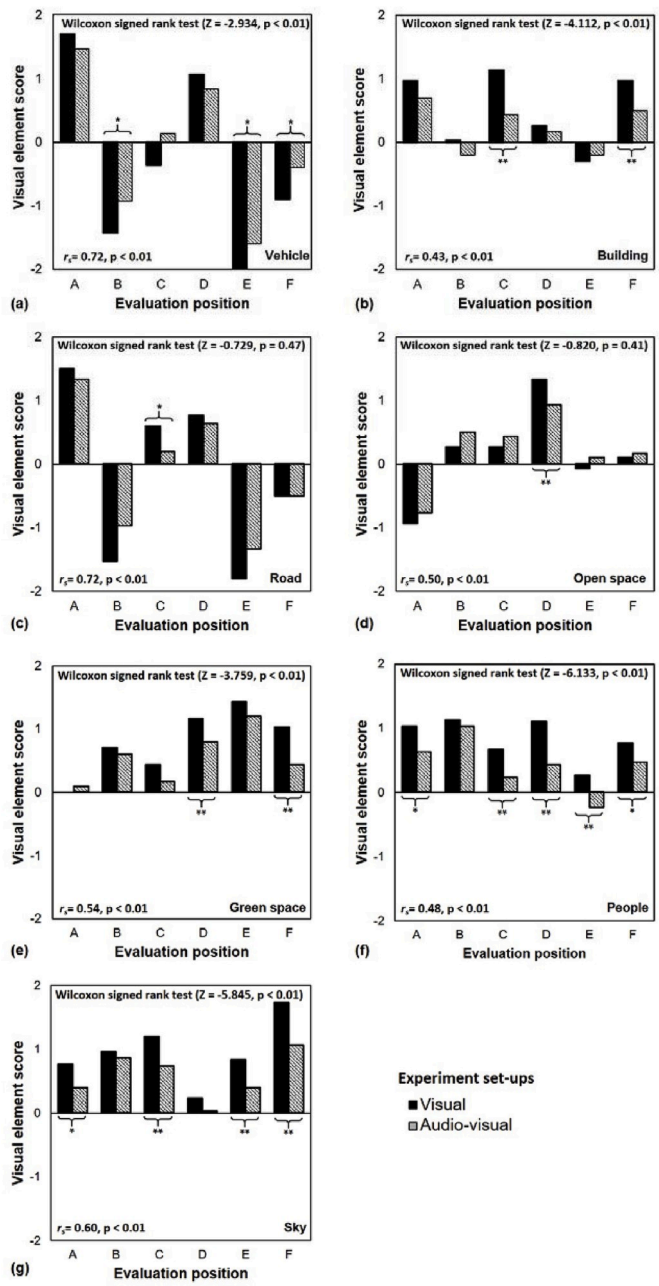

Fig. 3. Mean rating score with the perceived degree of visual elements according to the different experimental setups at the assessment sites: (a) vehicle, (b) building, (c) road, (d) open space, (e) green space, (f) people, and (g) sky; $^{*}p < .05$, $^{**}p < .01$.

in Fig. 4, the Spearman correlation coefficients (r_s) of the two different experiment setups ranged from 0.47 to 0.60, indicating that the response congruency of participants was quite high and significant ($p < .01$). "Satisfaction," "desire to stay," and feeling of "comfort" were evaluated higher in the green area (sites D and E) and recreational (site F) spaces. There was a low preference for site A, which was located close to a road. Spaces containing parks and those used for resting were positively assessed. Interestingly, the perception trends of the three soundscape qualities were similar for most of the assessed sites. Participants consistently assessed public squares (sites B and C) as producing positive emotions such as "satisfaction" and "comfort"; however, their "desire to stay" varied.

Places where the "traffic" sounds were dominantly perceived (Sites D and F) or the "vehicle" or "road" was visually attracting attention (Site D) were given higher preference compared to a park composed only of green area (Site E). This indicates that the urban context plays an important role in the perception of the soundscape of an urban environment. The soundscape quality preferences were generally lower in the audio-visual environment setup as compared with the only-visual environment: 0.16 ("satisfaction", $p < .05$), 0.15 ("desire to stay", $p < .05$), and 0.21 ("comfort", $p < .05$). The inappropriateness from the discrepancy between the expected degree and perceived degree of sound sources might have been the cause of the negative effect for assessment of urban soundscape quality.

4. Discussion

4.1. Influence of audio-visual interaction for soundscape quality enhancement

The correlations between the Spearman correlation coefficient (r_s) values were analyzed to investigate the audio-visual interactions within the urban environments, as presented in Table 3. First, regarding the "traffic" sounds among the expected sound sources, higher positive correlation values were observed with "vehicle" and "road" as compared to the others. This indicates that, in urban spaces, participants' expectation of "traffic" sounds is most significantly influenced by whether vehicles and roads are within the view of the spaces. Conversely, there was a weak negative correlation between "green spaces" and "sky" and a weak positive correlation between "building" and "people." It can be inferred that participants judged the "traffic" sounds as appropriate for the space when there was a large flow of population under the complex layout of the buildings. Therefore, in terms of design, urban spaces should be planned with minimum artificial elements such as vehicles and roads to reduce the level of expectation of traffic noises. Moreover, controlling the visual elements to reduce traffic noises and provide a tranquil environment to individuals will reduce negative chronic health effects such as cardiovascular problems and sleep disturbances [62] as well as bring positive psychological effects, including relaxation [63]. For example, it may be possible to reduce visual perception of an entire vehicle by controlling traffic through regulation or by controlling the speed of the vehicle. In addition, by planting street trees or creating small parks in urban spaces, we can draw attention to visual elements other than vehicles. The highest correlation of "human" sounds was with "people." In other words, the visual perception from the presence of people most significantly influenced the appropriateness of their sound. Owing to the numerous anthropological activities in shops, streets, and public squares, "human" sounds was positively correlated with "vehicle, " "building," "road," and "open space" elements. Finally, "birdsongs" yielded the highest positive correlation with "green space." Accordingly, the visual perception of "green spaces" significantly contributed to the subjective appropriateness for "birdsongs." Conversely, there was a negative correlation between "birdsongs" and artificial elements such as "vehicle" and "building," indicating that the visual perception of non-natural elements decreased the appropriateness of "birdsongs." In fact, it has been shown that birdsong loudness has a positive linear relationship with green areas in urban environments [64]. However, as shown in this study, it is possible to indirectly obtain soundscape quality enhancement—such as increase in soundscape pleasantness and eventfulness through natural sounds—simply through personal psychological effects in green areas, such as the appropriateness of birdsongs [65].

The mean difference (Δ, inappropriateness, the average score in the visual environment – the average score in the audio-visual environment) of audio-visual elements according to different experimental setups was also calculated, and the correlation between audio and visual elements was analyzed, as shown in Table 3. First, in urban environments, the visual attention to "vehicle" and "road" increased in spaces where "traffic" sounds were perceived more dominantly than participants expected to be appropriate for the space. Concerning "human" sounds, the incongruency between the degree of appropriateness based on

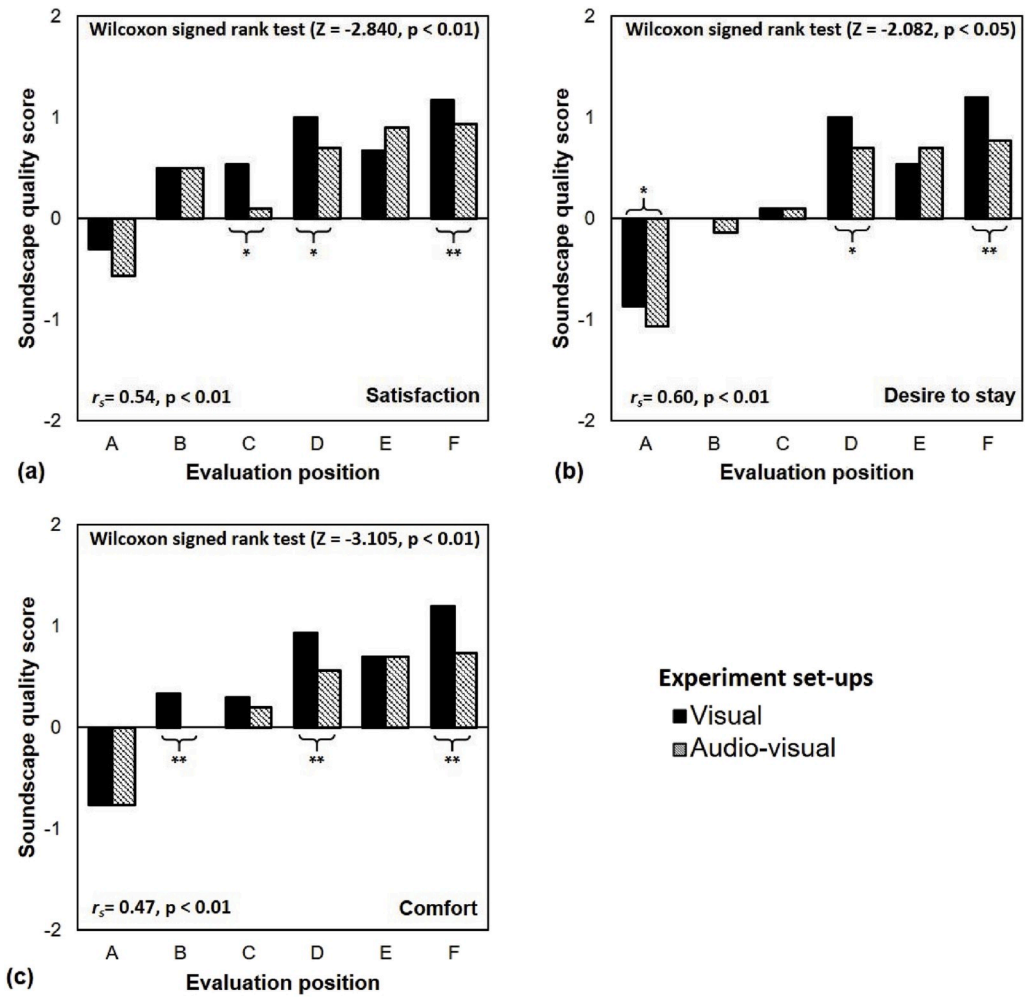

Fig. 4. Mean rating scores of soundscape quality according to the different experimental setups at the assessment sites: (a) satisfaction, (b) desire to stay, and (c) comfort; $*p < .05$, $**p < .01$.

Table 3
Spearman correlation coefficient (r_s) between the appropriateness of sound sources and visual elements.

	Traffic (Δ)		Human (Δ)		Birdsongs (Δ)	
Vehicle (Δ)	0.80 **	(0.38**)	0.32 **	(0.16*)	−0.37 **	(-0.02)
Building (Δ)	0.80 **	(-0.07)	0.32 **	(0.16*)	−0.18 *	(-0.12)
Road (Δ)	0.71 **	(0.23**)	0.32 **	(0.21**)	−0.31 **	(0.05)
Open space (Δ)	0.01	(0.09)	0.21 **	(0.19*)	0.15 *	(-0.10)
Green space (Δ)	−0.22 **	(-0.03)	0.06	(0.10)	0.46 **	(0.11)
People (Δ)	0.26 **	(0.15*)	0.39 **	(0.22**)	0.10	(0.01)
Sky (Δ)	−0.16 *	(-0.01)	0.13	(0.19**)	0.15 *	(-0.02)

participants' prior experience and the perceived degree changed the visual perception; specifically, participants concentrated more on the visual information related to "people" presence. It is noteworthy that a difference between expectation and perception of "human" sounds affected the recognition of all other visual elements, with the exception of "green space." Therefore, the visual quality of an urban environment can be improved considerably by inducing human activities in tranquil spaces or reducing human activities in overcrowded spaces. However, human sound, which is a dominant sound in an urban environment, has both a negative effect (for monumental space) and a positive effect (for park space) on soundscape quality depending on the location, so it is necessary take into account location characteristics [66,67]. In sum, the appropriateness of "traffic" and "human" sounds within a space significantly affects the visual perception of the space. "Birdsongs" did not exhibit any significant change regarding visual perception.

Values in parentheses are the r_s between the difference rating score (Δ) when changing from visual environment to audio-visual environments; $*p < .05$, $**p < .01$.

4.2. Effect of the appropriateness of the sound environment on soundscape quality

The correlations between the Spearman correlation coefficient (r_s) values were analyzed to investigate the effect of appropriateness of a sound environment on soundscape quality, as presented in Table 4. Participants' "satisfaction," "desire to stay," and "comfort" with the soundscape were negatively correlated with "traffic" and positively correlated with "birdsongs," and no significant correlation with "human" sounds was observed. Thus, participants' initial impression of the appropriateness of the soundscape quality in an urban area was more concerned with sounds owing to space functions and environmental characteristics such as "traffic" or natural sounds like "birdsongs" than with sounds produced by human activity.

The mean difference (inappropriateness) between participants'

Table 4
Spearman correlation coefficients (r_s) between the audio-visual elements and urban soundscape quality.

	Satisfaction (Δ)			Desire to stay (Δ)			Comfort (Δ)		
Sound sources									
Traffic (Δ)	−0.27	**	(0.08)	−0.25	**	(-0.08)	−0.29	**	(-0.06)
Human (Δ)	0.01		(0.04)	−0.01		(-0.01)	−0.01		(0.21**)
Birdsongs (Δ)	0.35	**	(0.09)	0.34	**	(0.12)	0.36	**	(0.08)
Visual elements									
Vehicle (Δ)	−0.21	**	(-0.04)	−0.19	*	(-0.11)	−0.29	**	(-0.07)
Building (Δ)	0.02		(0.09)	−0.07		(-0.01)	−0.10		(-0.14)
Road (Δ)	−0.18	*	(0.14)	−0.18	*	(-0.02)	−0.27	**	(-0.01)
Open space (Δ)	0.30	**	(-0.01)	0.36	**	(-0.04)	0.34	**	(0.09)
Green space (Δ)	0.26	**	(-0.05)	0.32	**	(-0.03)	0.32	**	(0.02)
People (Δ)	−0.07		(-0.15)	−0.01		(-0.01)	−0.04		(0.02)
Sky (Δ)	0.25	**	(0.05)	0.18	*	(-0.10)	0.23	**	(0.06)

rating score of each audio-visual element and soundscape quality in the visual and audio-visual environments was measured. Inappropriate "human" sounds were positively correlated with "comfort," and this significantly affected the urban soundscape quality. In other words, if "human" sounds were perceived more prominently than expected, the space was considered more comfortable. This is different from the results of previous studies that classified "human" sounds as unwanted sounds [68]. The results of this study, in contrast, indicate that the addition of an appropriate level of "human" sounds can enhance soundscape quality elements, such as comfort. Nonetheless, these observations are similar to those of one previous study [69], which found that the comfort of an urban space was higher when the crowd density was above a certain level (i.e., 0.10–0.25 persons/m^2).

As previously described, perception of sound sources is closely related to perceived visual elements in the context of audio-visual interactions. Further, visual perception is influenced by the difference between participants' judgment of appropriateness of sound sources and actual perceived degree of sound sources. Accordingly, it is necessary to determine whether a sound source has an independent influence on soundscape quality. The results revealed that "vehicle" and "road" were negatively correlated with the soundscape quality, whereas "open spaces," "green spaces," and "sky" were positively correlated with the soundscape quality. In addition, the changes in visual perception with the presence of sound information did not significantly influence the soundscape quality. Therefore, the effect of the appropriateness of the sound environment had a significantly independent effect on the assessment of soundscape quality from that of perceived visual elements.

Values in parentheses are the r_s between the difference rating score (Δ) when changing from visual environment to audio-visual environment; $^*p < .05$, $^{**}p < .01$.

4.3. Urban soundscape design guideline regarding the appropriateness of a sound environment

To design and improve an urban soundscape, we propose the following two measures related to the appropriateness of sound environments in the initial and late stages of urban soundscape design:

1) For the initial stage of soundscape design, the degree of appropriateness derived from participants' expectations of "traffic" sounds and "birdsongs" in an urban space is a critical factor that determines the initial perception of urban soundscape quality. The perception of visual elements such as "vehicle," "road," "open space," "green space," and "sky" is a vital design factor for shaping the early perception of urban soundscape quality. Moreover, the expectation of a sound environment significantly interacts with the perception of visual elements, which is directly related to each sound source—"traffic–vehicle," "human–people," and "birdsongs–green space." Therefore, in the early stages of designing the soundscapes of urban environments, it is important to have suitable audio-visual arrangements of design factors that are functionally related to transportation or to the natural environment. In other words, it is important to appropriately utilize natural sounds. which are a preferred sound type in urban spaces, to control "traffic" noise, as "traffic" noise negatively affects residents [62,63,70]. Birdsongs are considered the most preferred sound among natural sounds [6]. Therefore, they can be utilized to create the most effective masking effects in environments with lower sound levels of 52.5 dB or less, such as from traffic noise [71]. In addition, for water sounds among natural sounds, it may be possible to install fountains, streams, and water sculptures to provide a noise level that is close to ambient noise level or below 3 dB [72,73].
2) For the later stage of soundscape improvement, variations between the degree of appropriateness and perceived degrees of "traffic" and "birdsongs" did not change the soundscape quality; however, the variation for "human" sounds changed the soundscape quality. A positive change in "comfort" can be acquired when an urban environment is designed to increase "human" sounds to an appropriate level through infrastructure and design that foster human activity. Therefore, it can be said that people should be considered a design factor during the improvement stage of an urban environment, rather than during the planning stage, to enhance a soundscape. From a methodological perspective of urban environments, crowd density that affects the sound pressure level can be used as a design factor to set human sounds to an appropriate level [8]. Other ways of improvement include adjusting participants' duration of stay through providing music by installing hidden (invisible) speakers in open spaces [74] or organizing events that can appropriately create music-related or human-sound-related activities in a space [75].

5. Conclusion

In this study, the effect of the appropriateness of the sound environment on urban soundscape was investigated with audio-visual interaction aspects. Soundscape assessments were performed for various urban contexts. To ensure that the simulated experiences were similar to those at the assessment sites, VR technologies such as spatial audio and HMD were used. The soundscape quality was assessed in two experiment setups, and the relationship between the appropriateness of the sound environment and soundscape quality was investigated in detail. Finally, we proposed varied urban soundscape design guidelines regarding the appropriateness of the sound environment.

The application of the results presented in this paper are limited by the fact that all participants were in their 20s and all assessment sites were located in Seoul. In addition, owing to a lack of normality, the

extent of the influence of appropriateness on the perception of the soundscape quality could not be quantified. Therefore, to generalize the aforementioned influence on the proposed model for urban soundscape quality, further studies that consider a broader age range of participants and a wider spread of urban areas with more diverse functions are required. In addition, since the boundary of participants' responses was pre-selected, there is a possibility of bias in the experimental design. Therefore, future studies should facilitate free responses by participants, such as by including open-ended questions.

Despite the abovementioned limitations, we found that participants' appropriateness degree for "traffic" sounds and "birdsongs" were vital in determining the initial perception of the soundscape quality of urban environments. Further, to enhance "comfort" aspects of soundscape quality, "human" sounds originating from human activity are critical. In other words, during the initial stage of designing a new urban area, it is advisable to first consider topological and geographical elements such as roads, parks, and open outdoor spaces that are directly related to "traffic noise" and "birdsongs." At the later stage for improving the soundscape of a location, it is desirable to properly control human activities to induce "human" sounds at an appropriate level. Thus, these findings contribute to the design and development of comfortable and appealing urban soundscapes.

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Declarations of interest

There are no conflicts of interest to declare.

Declaration of competing interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome. We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

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